

Exam Date & Time: 10-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Agile Software Development for Engineers [IT-4308]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Is it possible to combine software process models for a given project? Justify with suitable example. (4)
 - A)
 - B) Illustrate the steps used to transfer from traditional approach to agile process by considering Retail and E-commerce Industry. (3)
 - C) Provide insights into how integrating DevOps into Agile workflows enhances the delivery pipeline and overall project success (3)
- 2) Illustrate with an example, how does site reliability engineering work? (4)
 - A)
 - B) Could you provide insight into "burndown charts" in project management? How are they utilized to track progress and manage work in kanban? (3)
 - C) Describe the characteristics of scrum framework. (3)
- 3) Illustrate the difference between CI and CD? (4)
 - A)
 - B) How does Continuous Integration (CI) ensure code quality and collaboration among team members? (3)
 - C) A company decides to transition from a traditional waterfall approach to Agile methodologies for their software development projects. Evaluate the challenges the organization might face during this transition and propose strategies to overcome these challenges while ensuring a smooth Agile adoption process. (3)
- 4) Consider a situation where an Agile team consistently delivers features on time but receives negative feedback from end-users regarding product usability and performance. Analyze the possible reasons behind this discrepancy between delivery speed and user satisfaction and propose approaches to incorporate user feedback more effectively into the development process. (4)
 - A)

- B) You're tasked with leading a retrospective meeting for an Agile team at the end of a sprint where several user stories were not completed. How would you facilitate this retrospective to identify the root causes of the incomplete work and formulate actionable improvements for the next sprint? (3)
- C) You're a Scrum Master for a distributed Agile team spanning multiple time zones. Describe how you would facilitate effective communication and collaboration among team members, ensuring alignment and synergy despite geographical barriers. (3)
- 5) How metrics like lead time, cycle time, and burndown charts can provide insights into team dynamics and productivity. (4)
- A)
- B) How the business involvement and team motivation managed in Agile product management? (3)
- C) Describe the possible combinations of pair programming in software development life cycle. (3)

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